

Module 9: Lambda Assignment

Problem Statement:

You work for XYZ Corporation. Your corporation wants to launch a new web-based application and they do not want their servers to be running all the time. It should also be managed by AWS. Implement suitable solutions.

Tasks To Be Performed:

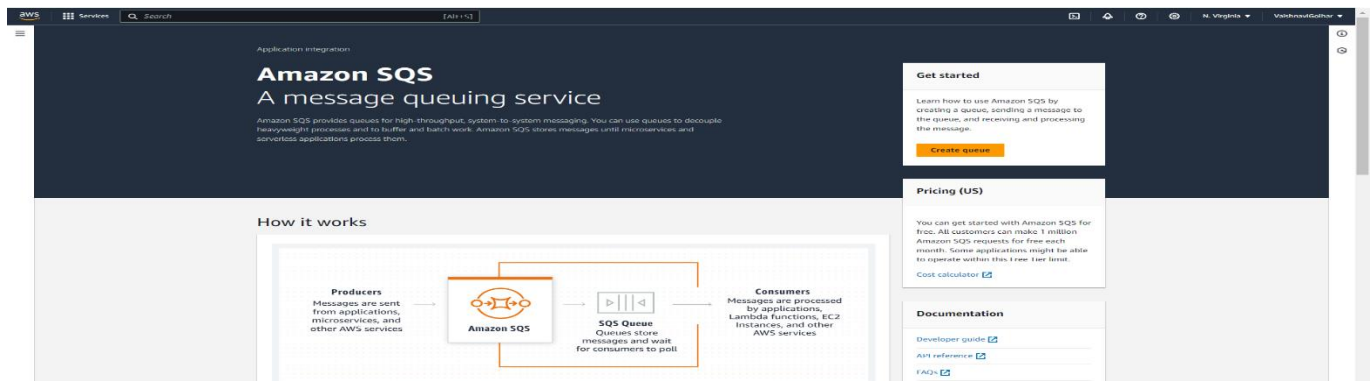
1. Create a sample Python Lambda function.
2. Set the Lambda Trigger as SQS and send a message to test invocations.

Solution:

1. Create an SQS Queue:

Go to the AWS Management Console and navigate to the SQS service.

Create a new SQS queue or choose an existing one.



Amazon SQS > Queues > Create queue

Create queue

Details

Type

Choose the queue type for your application or cloud infrastructure.

☒ **Standard** info

At least once delivery, message ordering isn't preserved

- At least once delivery
- Best-effort ordering

☐ **FIFO** info

Exactly-once delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

☐ You can't change the queue type after you create a queue.

Name

test-queue

A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration info

Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout info

30

Seconds

Should be between 0 seconds and 12 hours.

Message retention period info

4

Days

Should be between 1 minute and 14 days.

Delivery delay info

0

Seconds

Should be between 0 seconds and 15 minutes.

Maximum message size info

256

KiB

Should be between 1 KiB and 256 KiB.

Dead-letter queue - Optional info

Send undeliverable messages to a dead-letter queue.

Set this queue to receive undeliverable messages.

☒ Disabled

☐ Enabled

Tags - Optional info

A tag is a label assigned to an AWS resource. Use tags to search and filter your resources or track your AWS costs.

Key

Enter key

Value - optional

Enter value

Remove

Add new tag

You can add 49 more tags.

Cancel

Create queue

Queue test-queue created successfully

You can now send and receive messages.

Amazon SQS > Queues > test-queue

test-queue

EditDeletePurgeSend and receive messagesStart DLQ redrive

Details info

Name	test-queue	Type	Standard	ARN	arn:aws:sqs-east-1:161681788148:test-queue
Encryption	Amazon SQS key (SSE-SQS)	URL	https://sqs.us-east-1.amazonaws.com/161681788148/test-queue	Dead-letter queue	-

More

SNS subscriptions

Lambda triggers

EventBridge Pipes

Dead-letter queue

Monitoring

Tagging

Access policy

Encryption

Dead-letter queue redrive tasks

Subscription region

us-east-1

SNS subscriptions (0) info

Search subscriptions

Subscribe to Amazon SNS topic

Subscription ARN	Topic ARN
No Amazon SNS subscriptions	

2. Create a Lambda Function:

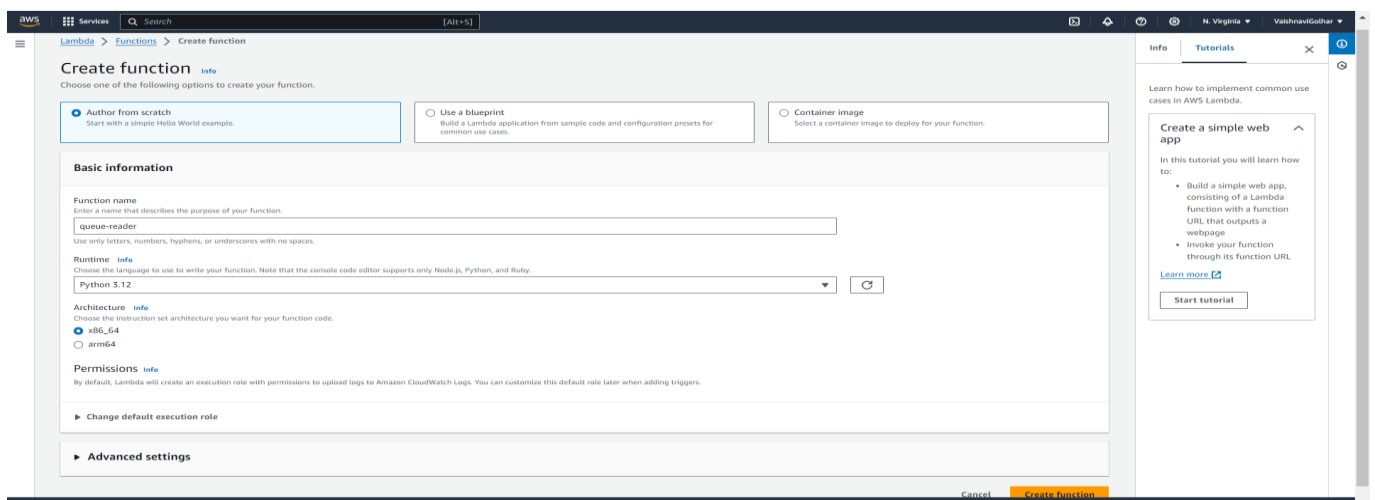
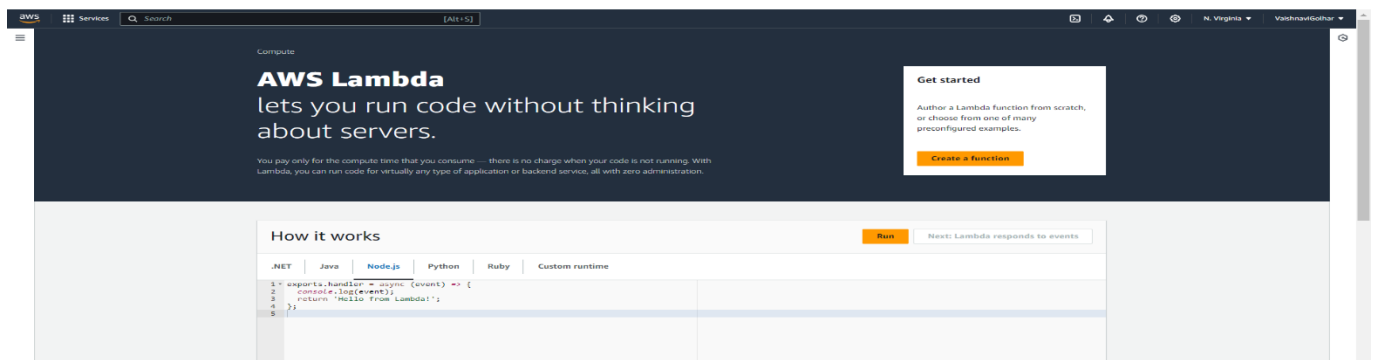
Go to the AWS Management Console and navigate to the Lambda service.

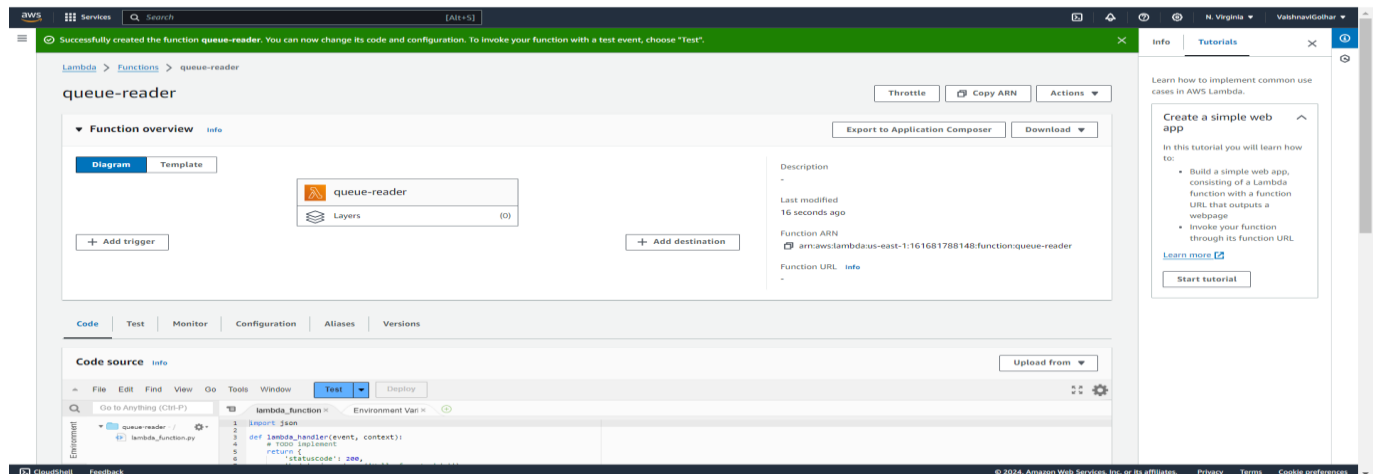
Click on "Create function".

Choose "Author from scratch".

Enter a name for your function, select Python as the runtime (or any other language you prefer), and choose an execution role.

Click on "Create function".



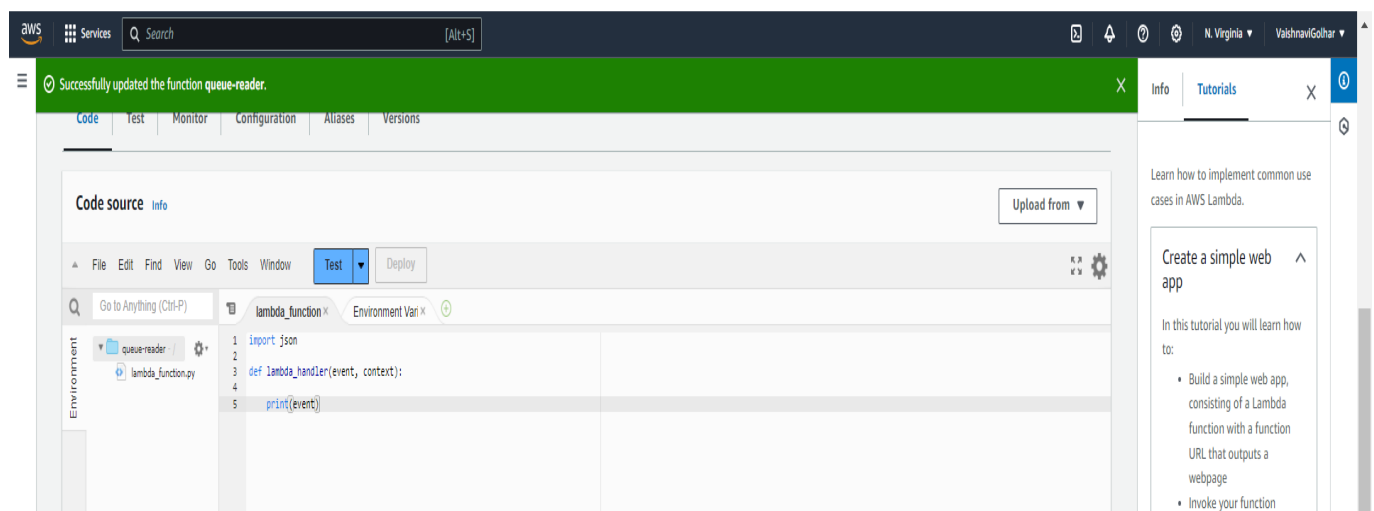


3. Write the Lambda Function Code:

In the function's configuration page, scroll down to the "Function code" section.

Write or paste your Python Lambda function code into the inline code editor.

Click on "Save" to save your function code



4. Set Up Lambda Trigger with SQS:

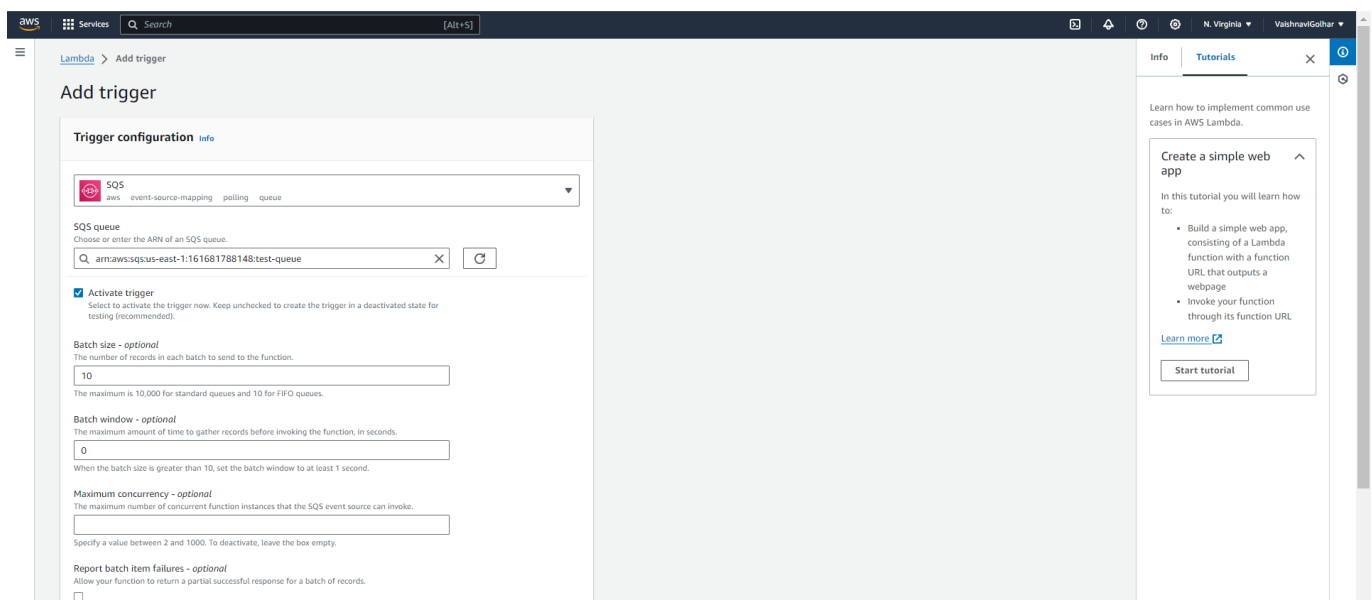
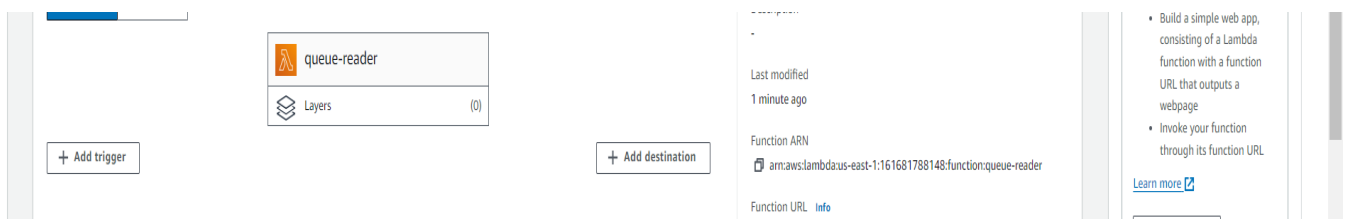
In the Lambda function configuration page, scroll down to the "Add trigger" section and click on it.

Choose "SQS" from the list of triggers.

Configure the trigger by selecting the SQS queue you want to use.

Set the batch size (the maximum number of messages to process in parallel).

Click on "Add" to add the trigger.



awsServicesSearch[Alt+S]

☰

Activate trigger
Select to activate the trigger now. Keep unchecked to create the trigger in a deactivated state for testing (recommended).

Batch size - optional

The number of records in each batch to send to the function.

10

The maximum is 10,000 for standard queues and 10 for FIFO queues.

Batch window - optional

The maximum amount of time to gather records before invoking the function, in seconds.

0

When the batch size is greater than 10, set the batch window to at least 1 second.

Maximum concurrency - optional

The maximum number of concurrent function instances that the SQS event source can invoke.

Specify a value between 2 and 1000. To deactivate, leave the box empty.

Report batch item failures - optional

Allow your function to return a partial successful response for a batch of records.

☐

Filter criteria - optional

Define the filtering criteria to determine whether or not to process an event. Each filter must be in a valid JSON format in filter rule syntax. Lambda processes an event if any one of the filters are met. [Learn more](#)

Add

During trigger creation, Lambda translates your filter(s) into a single JSON structure containing all your filtering criteria.

In order to read from the SQS trigger, your execution role must have proper permissions.

An error occurred when creating the trigger: The provided execution role does not have permissions to call ReceiveMessage on SQS

CancelAdd

InfoTutorials

Learn how to implement common use cases in AWS Lambda.

Create a simple web app

In this tutorial you will learn how to:

- Build a simple web app, consisting of a Lambda function with a function URL that outputs a webpage
- Invoke your function through its function URL

[Learn more](#)

Start tutorial

awsServicesSearch[Alt+S]

Identity and Access Management (IAM)

Search IAM

IAM

Roles

Roles (17) info

An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.

Search

RefreshDeleteCreate role

< 1 >

 [queue-reader-role-rx1ua3ex](#)

AWS Service: lambda

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies (SCPs)

Related consoles

[IAM Identity Center](#)

[AWS Organizations](#)

Summary

Creation date

March 22, 2024, 23:11 (UTC+05:30)

Last activity

-

ARN

arnawsiam:161681788148:role/service-role/queue-reader-role-rx1ua3ex

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Access Advisor

Revoke sessions

Permissions policies (1)

Info

You can attach up to 10 managed policies.

Filter by Type

All types

Search

Policy name

Type

Attached entities

☐

Policy name

☐

AWSLambdaBasicExecutionRole-e9fb8b89-09c2-4d20-b648-6f82b...

Customer managed

1

Permissions boundary (not set)

Generate policy based on CloudTrail events

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

Generate policy

No requests to generate a policy in the past 7 days.

[IAM](#) > [Roles](#) > [queue-reader-role-rx1ua3ex](#) > Add permissions

Attach policy to queue-reader-role-rx1ua3ex

Current permissions policies (1)

Other permissions policies (1/912)

Filter by Type

All types

Search

sqgs

3 matches

Policy name

Type

Description

☒

☐

AmazonSQSFullAccess

AWS managed

Provides full access to Amazon SQS via th...

Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

User groups

Users

Roles

Policies

Identity providers

Account settings

Access reports

Access Analyzer

External access

Unused access

Analyzer settings

Credential report

Organization activity

Service control policies (SCPs)

Related consoles

[IAM Identity Center](#)

[AWS Organizations](#)

Policy was successfully attached to role.

[IAM](#) > [Roles](#) > [queue-reader-role-rx1ua3ex](#)

queue-reader-role-rx1ua3ex

Info

Delete

Summary

Edit

Creation date

March 22, 2024, 23:11 (UTC+05:30)

Last activity

-

ARN

arnawsiam:161681788148:role/service-role/queue-reader-role-rx1ua3ex

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

Access Advisor

Revoke sessions

Permissions policies (2)

Info

You can attach up to 10 managed policies.

Filter by Type

All types

Search

Policy name

Type

Attached entities

☐

☒

AmazonSQSFullAccess

AWS managed

1

☐

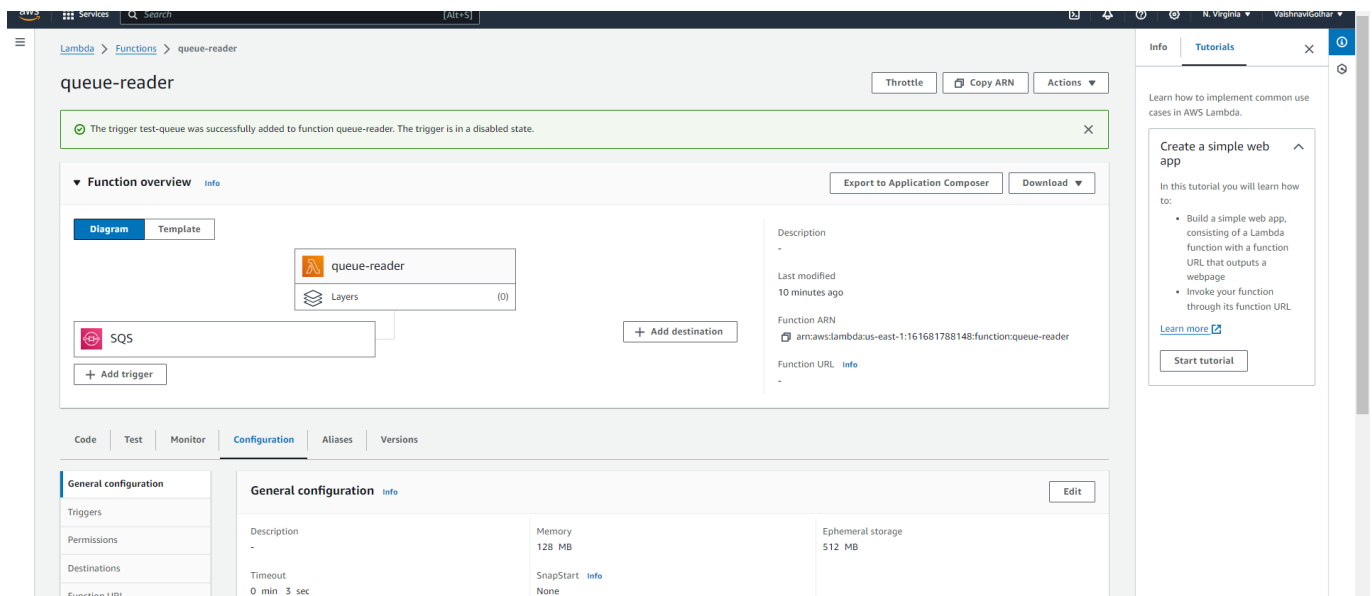
☒

AWSLambdaBasicExecutionRole-e9fb8b89-09c2-4d20-b648-6f82b...

Customer managed

1

Permissions boundary (not set)



5. Test the Configuration:

Go back to the SQS service in the AWS Management Console.

Select the SQS queue you configured as the trigger for your Lambda function.

Send a test message to the queue.

Monitor the Lambda function's execution logs in CloudWatch to see the output of the processing.

aws

Services

Search

[Alt+S]

N. Virginia

VaishnaviGolhar

Amazon SQS

Queues

test-queue

Send and receive messages

Send and receive messages

Send messages to and receive messages from a queue.

Send message

Info

Clear content

Send message

✔ Your message has been sent and is ready to be received.

View details

✕

Message body

Enter the message to send to the queue.

This is Lambda function assignment

Delivery delay

Info

0

Seconds

▼

Should be between 0 seconds and 15 minutes.

▶ Message attributes - Optional

Info

Receive messages

Info

Edit poll settings

Stop polling

Poll for messages

Messages available

0

Polling duration

30

Maximum message count

10

Polling progress

0 receives/second

0%

Messages (0)

View details

Delete

Search messages

<

1

>

ⓘ

aws

Services

Search

[Alt+S]

N. Virginia

VaishnaviGolhar

CloudWatch

×

Favorites and recents

▶

Dashboards

▼ Alarms

0

0

0

1

In alarm

All alarms

Billing

▼ Logs

Log groups

Log Anomalies

Live Tail

CloudWatch

>

Log groups

Log groups (1)

By default, we only load up to 10000 log groups.

Refresh

Actions

View in Logs Insights

Start tailing

Create log group

Filter log groups or try prefix search

Exact match

<

1

>

ⓘ

☐

Log group

▼

☐

/aws/lambda/queue-reader

▼

Log class

▼

Standard

▼

Anomaly d...

▼

Configure

▼

Data protection

▼

-

▼

Sensitive data co...

▼

-

▼

Retention

▼

Never expire

▼

Metric filters

▼

-

▼

Contributor insig...

CloudWatch

Favorites and recents

Dashboards

Alarms

0

0

0

1

In alarm

All alarms

Billing

Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Metrics

X-Ray traces

Events

Application Signals

Network monitoring

Internet Monitor

Network Monitor

Insights

Settings

Getting Started

What's new

CloudWatch > Log groups > /aws/lambda/queue-reader

Actions

View in Logs Insights

Start tailing

Search log group

Log group details

Log class

Standard

Info

ARN

arn:aws:logs:us-east-1:161681788148:log-group:/aws/lambda/queue-reader:

Creation time

2 minutes ago

Retention

Never expire

Stored bytes

-

Metric filters

0

Subscription filters

0

Contributor Insights rules

-

KMS key ID

-

Anomaly detection

Configure

Data protection

-

Sensitive data count

-

Log streams

Tags

Anomaly detection

Metric filters

Subscription filters

Contributor Insights

Data protection

Log streams (1)

Filter log streams or try prefix search

Exact match

Show expired

Info

1

>

0

Log stream

Last event time

2024/03/22/[SLATEST]69af45606df3463eb98bb28e097f3b80

2024-03-22 23:25:40 (UTC+05:30)

CloudWatch

Favorites and recents

Dashboards

Alarms

0

0

0

1

In alarm

All alarms

Billing

Logs

Log groups

Log Anomalies

Live Tail

Logs Insights

Metrics

X-Ray traces

Events

Application Signals

Network monitoring

Internet Monitor

Network Monitor

CloudWatch > Log groups > /aws/lambda/queue-reader > 2024/03/22/[SLATEST]69af45606df3463eb98bb28e097f3b80

Actions

Start tailing

Create metric filter

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events

Clear

1m

30m

1h

12h

Custom

Local timezone

Display

Timestamp

Message

No older events at this moment. [Retry](#)

2024-03-22T23:25:40.013+05:30

INIT_START Runtime Version: python:3.12.v21 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:0087788f33e3d6b95524220734bb2b31388197821920f0844d089551d6fa015

2024-03-22T23:25:40.102+05:30

START RequestId: 1022095c-4698-5811-956a-532b0edc37d7 Version: SLATEST

2024-03-22T23:25:40.103+05:30

{'Records': [{'messageId': 'f4b18eb2-0866-455d-9fef-20999430ff08', 'receiptHandle': 'AQEBnJ8uHv00ZKI0yYSIQ4jLX2H9PUuSVJ24eushyXNBYSdPLHxJPV014Ff8Dpgu1m7LJ892c18uJlW+IDfWStoFDl0jQ6sqE3LyD/PiAGVOyVfghxatLob35SMjK3JlU7OKLCP0uThiTEps+ygBUC57Q3wWw8KitH4HRU0jSTXtWkZXGGSISatgmbJ4AEI/ZxPH+ELL3GL7A0eSnQTzgf+GjUirniLowq2llyxsaetCJTQ5jurs00vk10z6yKozJ0WmSmJ0ZVvPrCp+m/B8SukCux+7890yEAF1RjBUvHjX2cFiuVjSkYjYOK+QIBrSzuq4LS2AMWU4F4St6nqOPuqLymj/cuAhZc8B2Xav3', 'body': 'This is Lambda function assignment', 'attributes': {'ApproximateReceiveCount': '1', 'SentTimestamp': '1711138139851', 'SenderId': '161681788148', 'ApproximateFirstReceiveTimestamp': '1711138139856', 'messageAttributes': {}, 'md5OfBody': '66e16d9faad20b11079f3d336a3697', 'eventSource': 'aws:sqs', 'eventSourceARN': 'arn:aws:sqs:us-east-1:161681788148:test-queue', 'awsRegion': 'us-east-1'}]}}

2024-03-22T23:25:40.104+05:30

END RequestId: 1022095c-4698-5811-956a-532b0edc37d7

2024-03-22T23:25:40.104+05:30

REPORT RequestId: 1022095c-4698-5811-956a-532b0edc37d7 Duration: 2.41 ms Billed Duration: 3 ms Memory Size: 128 MB Max Memory Used: 34 MB Init Duration: 88.26 ms

No newer events at this moment. [Auto retry paused. Resume](#)

Log events

Actions ▾

Start tailing

Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q Filter events

Clear1m30m1h12hCustomLocal timezoneDisplay

Timestamp	Message
No older events at this moment. Retry	
2024-03-22T23:25:40.013+05:30	INIT_START Runtime Version: python:3.12.v21 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:0087788f33e3d6b9552242d7340b2b31308197021920f044d089552d6fa015
2024-03-22T23:25:40.102+05:30	START RequestId: 10220d5c-4690-5811-956a-532b8edc37d7 Version: \$LATEST
2024-03-22T23:25:40.103+05:30	{'Records': [{'messageId': 'f4b18eb2-0a66-455d-9fef-2099d43bff88', 'receiptHandle': 'AQEMjJaUHV0BZXI0yYSIGQ4jLX2H9PUUSVJ14eushyX0BvYdSpLHGxJPVD14FFdDpgvU1Rm7L3892c18uUuR+MDfWvStoF0jQ6sqKE3LyD/P1aG0YOy... {'Records': [{'messageId': 'f4b18eb2-0a66-455d-9fef-2099d43bff88', 'receiptHandle': 'AQEMjJaUHV0BZXI0yYSIGQ4jLX2H9PUUSVJ14eushyX0BvYdSpLHGxJPVD14FFdDpgvU1Rm7L3892c18uUuR+MDfWvStoF0jQ6sqKE3LyD/P1aG0YOyQTzpf+GjUuRi0hYq2LjgXsaetCjTQ5jur9d0vK10z6RyN0zj30WMSmJ0Z1VPrCp+m/885UKuX+7890YeAF1r3BuVHjX2cFwUwJSkyjKOK+QlBRsZAlqAL524mInuR4Fhst6nQPUQlYmmj/ouAhZz8B2XAV3', 'body': 'This is Lambda function assignment', 'attributes': {'ApproximateReceiveCount': '1', 'SentTimestamp': '1711130139851', 'SenderId': '161681788148', 'ApproximateFirstReceiveTimestamp': '1711130139856'}, 'messageAttributes': {}, 'md5OfBody': '66e16ddfaad20b221079ff3d336a3697', 'eventSource': 'aws:sqs', 'eventSourceARN': 'arn:aws:sqs:us-east-1:161681788148:test-queue', 'awsRegion': 'us-east-1'}}]} Copy
2024-03-22T23:25:40.104+05:30	END RequestId: 10220d5c-4690-5811-956a-532b8edc37d7
2024-03-22T23:25:40.104+05:30	REPORT RequestId: 10220d5c-4690-5811-956a-532b8edc37d7 Duration: 2.41 ms Billed Duration: 3 ms Memory Size: 128 MB Max Memory Used: 34 MB Init Duration: 88.26 ms
No newer events at this moment. Auto retry paused Resume	